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A machine learning framework for soft skills assessment: Leveraging serious games in higher education

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ABSTRACT

This study explores the use of serious games combined with machine learning techniques to evaluate student efficiency and determine appropriate degree programs. The main research question addressed is: How can predictive machine learning algorithms, applied to soft skills data collected through serious games, identify the most suitable study path for each student, thereby improving academic orientation and enhancing the likelihood of educational success? The research involved 211 university students from a single university in Italy, focusing on the application of predictive algorithms based on participants' soft skills and academic performance (GPA). The study employs logistic regression models to classify students into three degree programs: Economic Business Administration (EBA), Sports and Motor Activity Sciences (SMAS), and Modern Literature (ML). Subsequent analysis using ANOVA Kruskal-Wallis identified significant differences in predictive results across various demographic and behavioral subgroups.

As an additional experiment, Data Envelopment Analysis (DEA) was employed to further understand student efficiency. DEA used soft skills as inputs and GPA as the output to calculate efficiency scores, which were then analyzed for all groups. The results highlighted significant differences in efficiency scores among different courses, age groups, and genders, but not among different technology groups or daily screen time.

The findings demonstrate that serious games represent a breakthrough innovation in educational assessment, providing a highly effective platform for capturing soft skills through engaging behavioral scenarios that traditional methods cannot access. This game-based approach, when integrated with machine learning algorithms, successfully classifies students based on comprehensive competency profiles, offering a revolutionary advancement in personalized academic orientation. The serious games methodology proves superior to conventional assessment techniques by enabling real-time collection of rich behavioral data that reveals authentic student capabilities beyond academic performance metrics. Supplementary DEA analysis confirmed these findings. Despite limitations such as sample size and focus on specific devices, this study establishes serious games as a transformative educational technology that fundamentally enhances academic orientation strategies. Future research should explore larger and more diverse samples, integrate advanced technologies, and conduct longitudinal studies to further validate and enhance the serious games approach in educational contexts.

1. Introduction

The past few decades have seen a rapid utilization of serious games for learning, education, and skill development. The concept of gamification has been applied in the context of education for the development of serious games to provide improved engagement (Maedche, 2017), enhanced teaching and learning experiences (Checa & Bustillo, 2020; Gomez & Suarez, 2021), productivity, cognition, and satisfaction (Wouters et al., 2013), as well as behavior modification (Rodrigues et al., 2014).

Serious games can be defined as digital games designed primarily for purposes other than pure entertainment. They are intentionally developed to achieve educational, training, or behavioral objectives. These games incorporate learning mechanisms, challenges, and feedback systems that support the acquisition of knowledge and the development of a wide range of competences, including transversal skills. Such skills include critical thinking, communication, adaptability, emotional regulation, problem-solving, and teamwork, all of which are

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increasingly recognized as essential in both academic and professional settings. Helping students to identify the most suitable academic path is a key component of effective educational guidance. While academic performance is often used as the main indicator, it does not always reflect a student's full potential or align with their personal aptitudes. Transversal skills, which are typically developed over time through varied learning experiences, play a crucial role in shaping students' ability to succeed in different educational contexts. By understanding and measuring these skills, educators and institutions can better support students in making informed and individualized academic decisions. In light of the COVID-19 pandemic, the use of screens for educational purposes has gained significant attention. A recent study explored the impacts of screen use during the pandemic on children and adolescents, providing valuable insights into how serious games could be adapted to these new conditions (Resende et al., 2024). Moreover, the application of serious games is not limited to general education but has been explored in specialized fields such as chemical engineering (Díaz et al., 2024). Other studies have investigated the implementation of serious games in specific educational contexts, including nursing education (Sánchez-Valdeón et al., 2023), computer science education (Lampropoulos et al., 2023), project management (Hellström et al., 2023), and entrepreneurship education (Martins et al., 2023).

In this respect, serious games have been able to leverage the positive effects of gamification for learning and skill development, while at the same time, mitigating the serious negative implications of traditional computer games (Connolly et al., 2012). This marks the evolution of traditional computer games from purely entertainment purposes to the context of educational learning and skill development (Baptista & Oliveira, 2019).

On one hand, different serious games have been developed that cater to nurse education (Min et al., 2022), medical education (Gorbanev et al., 2018; Haoran et al., 2019), STEM education (Boyle et al., 2016), dental education (Sipiyaruk et al., 2021), and health education (Sharifzadeh et al., 2020). On the other hand, the use and deployment of serious games are linked with improvements in strategic thinking, cognitive skills, decision-making skills (Clark et al., 2016; Gomez & Suarez, 2021), multi-tasking, communication skills, analytical skills, clinical reasoning, self-efficacy (Bayram & Caliskan, 2019; Tan et al., 2016), software project management skills (Calderon & Ruiz, 2015), and collaborative learning (Tsoy et al., 2019).

Despite the considerable recent developments for the application of serious games in the educational setting, there is still a lack of consensus regarding their overall efficacy and utility, with various studies questioning the validity and rigor of existing findings (Hays, 2005; Hanus & Fox, 2015; A. et al., 2016; Petri & von Wangenheim, 2017). It is for this reason that further examination is needed regarding the effectiveness of serious games not only in skill development but also in their potential to support academic guidance through soft skill analysis. In this study, the application of machine learning with predictive algorithms, based on participants' soft skills and academic performance, will provide valuable insights to improve academic orientation and enhance students' chances of success in their chosen fields of study. The analysis of the collected data will allow for the development of accurate models that identify the most suitable degree program for each participant, considering both academic performance and transversal skills.

The work will be structured into five main sections. Following the introduction, there will be a background section discussing related studies and outlining the research question. Subsequently, the methodology will describe the data collection process, preprocessing, participant information, and analysis techniques used. The results section will provide a comprehensive discussion, establishing the significance of the findings in relation to existing studies. Finally, the conclusion will present recommendations for future research in this field.

2. Literature review

2.1. Serious games in education

A vast body of literature has explored the potential of serious games and their applications across various domains, including education and skill development (Checa & Bustillo, 2020; Gomez & Suarez, 2021), healthcare (Graafland et al., 2014; Harrington et al., 2018), and professional environments (Joshi et al., 2021; Shi et al., 2019). In the educational field, serious games have been employed to enrich learning experiences in diverse disciplines such as pharmacology (Lancaster, 2014), medical training (Luctkar-Flude et al., 2021), sex education (Arnab et al., 2013), communication studies (Guillen-Nieto & Aleson-Carbonell, 2012; Hanus & Fox, 2015), computer networking (Minovic et al., 2015), media studies (Katsaounidou et al., 2019), and urban design and architecture (Fonseca et al., 2021). Recent developments have explored the integration of virtual reality (Fonseca et al., 2021; Lancaster, 2014) and mobile-based platforms (Sánchez & Olivares, 2011), including innovative applications in relationship and sex education for school students (Arnab et al., 2013). Beyond traditional educational contexts, serious games have proven to be valuable tools for professional training and soft skill development, in response to the rapidly evolving demands of the labor market (Romero et al., 2015). In industrial settings, serious games have been used for safety training, such as immersive virtual reality for fall prevention in construction (Shi et al., 2019), biofeedback-based simulators for risk assessment (Habibnezhad et al., 2021), and virtual reality applications in the precast/pre-stressed cement industry (Joshi et al., 2021). Some studies have reported mixed results when comparing VR to traditional training methods like PowerPoint presentations (Leder et al., 2019). In the aviation sector, mobile-based VR platforms have improved risk awareness (Chittaro et al., 2018), while immersive VR-based serious games have demonstrated higher knowledge retention and engagement compared to traditional safety card methods (L. & F., 2015). Serious games are now widely recognized as effective tools for both the development and assessment of transversal skills, due to their capacity to simulate authentic scenarios, foster active student engagement, and generate rich behavioral data (Bellotti et al., 2013). When specifically designed to train soft skills, these learning environments show high feasibility and sustained user engagement, offering structured contexts suitable for the observation and evaluation of individual competencies during gameplay (Altomari et al., 2023). Game-based learning experiences have been associated with significant improvements in transversal competencies, particularly interpersonal and intrapersonal abilities, emotional regulation, communication, and critical thinking (Novoseltseva et al., 2022). In higher education, the integration of gameplay data into structured evaluation frameworks enables deeper understanding of learning processes, overcoming many of the limitations of traditional assessment approaches (Pitic & Irimiaş, 2023).

2.2. Applications of machine learning in serious games

Serious games represent an innovative application of machine learning techniques, offering a unique opportunity to personalize the learning experience and improve educational effectiveness. By integrating advanced algorithms, it is possible to dynamically adapt the game's difficulty level, monitor student performance in real-time, and provide immediate and personalized feedback. A significant example is the integration of modern artificial intelligence in serious games to create personalized and adaptive gaming experiences. This approach allows the game's difficulty level to be dynamically adjusted based on the player's abilities, thereby enhancing engagement and learning effectiveness (Streicher & Smeddinck, 2016). A generic application for stealth assessment in serious games highlights how reusable AI components can be implemented to improve educational effectiveness. The reusability of AI components facilitates the implementation of advanced machine learning techniques, making serious games more efficient and scalable (Westera et al., 2020). The use of decision-making techniques based on artificial intelligence in serious games demonstrates how integrating machine learning into decision-making processes within games can significantly improve students' problem-solving and decision-making skills (M. Frutos-Pascual & B. G. Zapirain, 2017). Multimodal affective state recognition in serious games, using deep learning methods, allows the game to be adapted to the player's emotions. This enhancement of engagement and motivation during learning creates a more engaging gaming experience (A. Psaltis et al., 2016). The analysis of different learning levels in serious games examines how these tools can facilitate meaningful and lasting learning. Understanding the significance and deep learning in serious games offers valuable insights for improving educational design and effectiveness (Mitgutsch, 2011). Through the analysis of behavioral data collected during gameplay, machine learning algorithms can identify cognitive anomalies and predict players' memory capacities (Lima et al., 2024). Machine learning, applied through a logistic regression approach on a serious game dataset, has demonstrated high accuracy in distinguishing between children with and without Neurodevelopmental Disorders, highlighting new opportunities for early diagnosis and personalized interventions (Toki et al., 2024).

Moreover, machine learning techniques applied to behavioral logs captured during gameplay have proven particularly effective in predicting academic performance, offering a more robust and adaptive alternative to conventional statistical approaches (Daoudi et al., 2021; Loh et al., 2015). The incorporation of intelligent virtual agents further enriches serious game environments, facilitating the creation of interactive educational scenarios and enabling structured behavioral observation, particularly in school-based soft skills development (Dell'Aquila et al., 2022). In parallel, advanced analysis of gameplay data using machine learning has enabled accurate prediction of study trajectories, motivational patterns, and interaction strategies, thereby contributing to the advancement of intelligent, data-informed learning environments (Medina-Merodio et al., 2024).

In this study, we apply predictive machine learning algorithms to predict students' degree programs based on their soft skills, measured through a serious game using the dimensions Priority, Planning and Time, and their academic performance (GPA). The degree programs are Economic Business Administration (EBA), Sport and Motor Activity Sciences (SMAS), Modern Literature (ML). Subsequently, we will analyze the results using ANOVA across various subgroups (technological media used, age, gender, year of study, daily screen time in hours). The use of predictive machine learning algorithms will enable the development of accurate models to predict the most suitable academic path for each participant, based not only on their academic performance but also on their soft skills, which are often crucial for success in various professional fields. ANOVA analysis will provide further insights, allowing us to identify significant differences between student subgroups in terms of competencies and academic success. This type of analysis will help us develop personalized interventions and targeted educational strategies, improving academic orientation and enhancing the likelihood of students' success in their chosen fields of study. Ultimately, this study will not only contribute to a better understanding of the role of soft skills and academic performance in the choice of a degree program but also offer new perspectives on the application of machine learning in education.

2.3. *Academic guidance research gaps: evidence for ML and serious games*

The increasing complexity and diversity of academic pathways has made it progressively more difficult for students to make informed educational choices, while traditional guidance systems reveal structural and functional limitations that highlight the need for innovative technological approaches. It has been shown that conventional academic advising methods fail to adequately meet the needs of contemporary students, with effectiveness rates falling below 45.6 % and systemic scalability issues across institutions (Jaradat & Mustafa, 2017). Educational decisions are now understood as complex and often non-linear processes, requiring more sophisticated models. Using network analysis, researchers have demonstrated that individual learning paths exhibit bifurcations, while collective trajectories reveal fractal properties, showing a strong correlation between online learning rates and academic performance ($r = 0.754$, $p = 0.012$), with deep learning models achieving 94 % accuracy in outcome prediction based on path complexity (Ortiz-Vilchis & Ramirez-Arellano, 2023). Beyond the mathematical and structural dimensions, psychological, socioeconomic, and cultural factors further contribute to the intricacy of academic decision-making. It has been found that student choices are simultaneously influenced by personal traits, aspirations, socio-psychological identities, emotional responses, and risk calculations, exacerbated by a "social stratification of information" that disadvantages students from underprivileged backgrounds. Despite broader educational offerings, part-time enrollments declined by 61 % between 2010 and 2017, indicating deteriorating access conditions for working-class students and the inability of support systems to address differentiated needs (Callender & Dougherty et al., 2018). Quantitative evidence further confirms the systematic inefficacy of traditional methods. A correlational study revealed that 45.6 % of students stated that academic advisors did not influence their degree choice, and 53.2 % reported not receiving any significant guidance during high school. Average effectiveness scores were low (2.30/5.0 for university advisors; 2.08/5.0 for high school counselors), and no significant correlation was found between academic guidance and persistence in degree programs, with 36.9 % of students changing major by the second year despite having received support (Jaradat & Mustafa, 2017). Scalability and quality issues further underscore structural shortcomings, with a widespread lack of available counselors, difficulty in scheduling meetings, and high institutional costs for quality support platforms. These are compounded by inconsistent advice, inaccurate information, and service quality varying significantly between institutions (Akiba & Fraboni, 2023). A systematic review confirmed that advisors are often overburdened and have limited time per student, leading to widespread dissatisfaction (Iatrellis et al., 2017). In this context, the application of machine learning emerges as a promising solution. It has been emphasized that current guidance practices continue to rely on traditional group sessions that fail to provide personalized attention. A model based on Huber Regressor achieved 93.06 % accuracy in personalized recommendations (Majjate et al., 2024). Additionally, research has shown that standardized approaches compromise learning quality, while AI-powered personalization increases engagement, motivation, and knowledge retention (Gligorea et al., 2023). Further opportunities lie in the use of serious games, whose application in academic guidance remains limited. A review of the literature revealed that only 2.9 % of available studies focus on technological applications for guidance, despite 91.4 % of them reporting positive outcomes in educational game use (Tene et al., 2025). These tools, through interactive and immersive technologies, offer immediate feedback, personalized learning paths, and deep cognitive engagement, yet their potential in academic advising remains largely unexplored (Facchino et al., 2025).

In summary, the body of evidence clearly demonstrates that modern academic pathways are too complex to be addressed by linear, one-size-fits-all tools. The structural complexity, informational inequalities, institutional limitations, and emerging technological capabilities together call for the development of a new generation of academic guidance tools—personalized, scalable, and effective.

2.4. *Research question*

The increasing complexity and diversity of academic pathways necessitate advanced tools to support students in selecting the most suitable course of study. Traditionally, academic orientation has relied on standardized criteria that often fail to fully account for the unique skills and interests of individual students. This research introduces the question of how predictive machine learning algorithms can be used to more accurately identify the ideal study path for each student.

Research Question: *How can predictive machine learning algorithms, applied to soft skills data collected through serious games, identify the most suitable study path for each student, thereby improving academic orientation and enhancing the likelihood of educational success?*

The use of serious games to collect data on soft skills, combined with academic performance, provides a unique opportunity to apply advanced predictive models. These models can uncover hidden patterns and relationships among variables, offering personalized recommendations that significantly enhance academic orientation.

The application of machine learning in this context aims to:

1. Personalize Academic Orientation: Leverage detailed data collected from serious games to provide personalized recommendations that consider the specific skills and interests of students.
2. Increase Educational Success: Use predictive models to improve the accuracy of guidance provided to students, thereby increasing their chances of success in their chosen study paths.
3. Optimize the Use of Educational Technologies: Integrate the potential of serious games and machine learning to transform academic orientation, making it more effective and evidence-based.

This research will not only contribute to improving academic orientation but will also offer new insights into the use of machine learning and serious games in education, paving the way for more personalized and data-driven educational interventions.

3. Methodology

After a brief overview of the serious game administered to the students, we will apply the logistic regression algorithm in a machine learning environment to the features identified by the serious game (soft skills and GPA), using the categorical variable "course of study" as the target. The results will then be analyzed by groups (technological media used, age, gender, technology used, daily screen time in hours, year of study). This process will allow us to systematically and accurately identify and evaluate the skills developed by the participants.

Through the analysis of the collected data, it will be possible to gain an in-depth understanding of the acquired soft skills, highlighting areas of strength and those that need further improvement. This approach will enable the improvement of the training process and optimize the effectiveness of the serious game in the education of students.

3.1. Analysis workflow and application context

Analyses were performed using Orange Data Mining v3.36 on an Apple M1 Pro system with 16 GB RAM and 1 TB storage, running macOS Sonoma 14.2.1. This configuration, combined with the implementation of advanced machine learning techniques, ensured efficiency and reproducibility of our analyses. The importance of such machine learning methodologies in extracting meaningful insights and predictive models from complex data sets has been previously emphasized and validated in similar studies within the field of public health performance assessment (Santamato et al., 2023), especially in the context of digital healthcare transformation, where machine learning supports high-precision risk prediction and intelligent diagnostic systems across various domains (Iacoviello et al., 2025; Marengo et al., 2024). Fig. 1 represents the methodological workflow used in the Orange software for analyzing the data collected through the serious game. This workflow is divided into several key phases, each highlighted with colored boxes for better visual comprehension.

In the Data Preprocessing phase, highlighted by the orange box, the necessary steps are taken to prepare the raw data for analysis, ensuring it is clean and properly formatted. This includes selecting relevant columns, transforming data, and handling outliers. The blue box represents the Training and Testing Split phase, where the dataset is divided into two parts, with one portion designated for training the model and the other for testing. This step is crucial to ensure the validity of the model's results. The Evaluation of the Model's Performance phase, highlighted by the light blue box, involves applying various machine learning algorithms to compare their performances and select the best model. Algorithms such as Naive Bayes, SVM, and Random Forest are tested and evaluated in this phase. The green box highlights the ANOVA Analysis phase, which includes statistical analysis to identify significant differences among the various subgroups defined in the dataset. This analysis helps to better understand data variations and the effectiveness of predictive models. Finally, the Visual Representation phase, outlined by the red box, creates visual representations of the analysis results. Tools such as the confusion matrix and ROC curve are used to facilitate the interpretation and communication of the findings, allowing for a clear and immediate understanding of the model's performance.

This methodological workflow, with its well-defined and visually

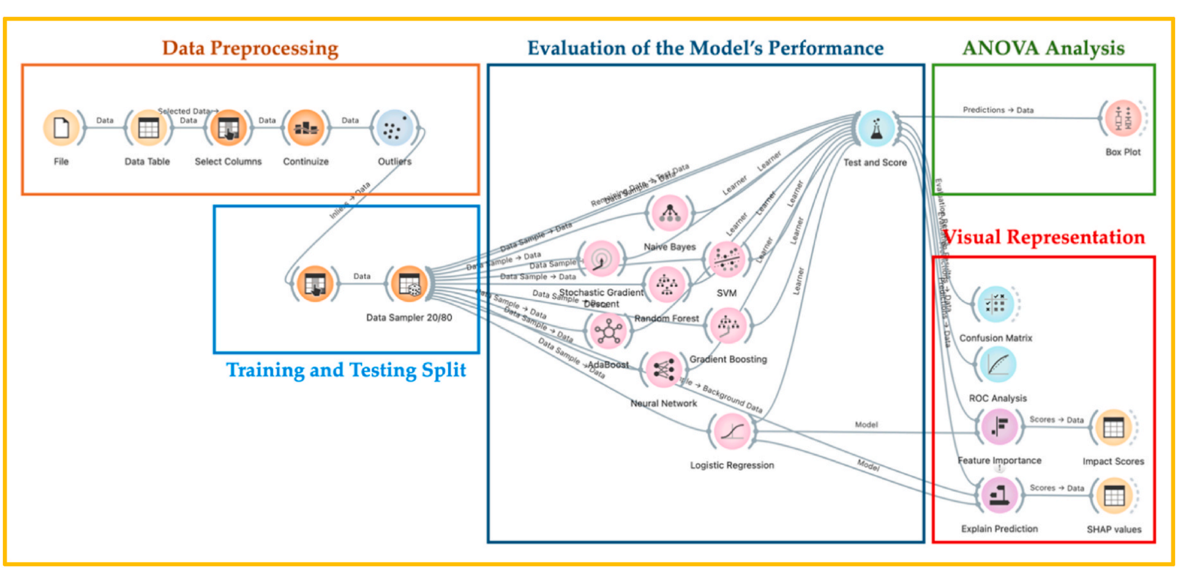

Fig. 1. Methodological workflow used in the Orange software.

distinct phases, enables systematic and thorough analysis of the data collected through the serious game, ensuring the reliability and reproducibility of the results.

3.2. Participants and data collection

A total of 211 students participated in the study. The descriptive statistics for the subgroups (technological media used, age, gender, year of study, daily screen time in hours) as expressed by the administered tests are reported in Table 1. The sample used for this study was a convenience sample, drawn from a population of university students enrolled in computer science courses across three different degree programs, which was chosen as the target variable for the predictive model. The choice of the degree program as the target variable is based on its relevance in determining the specificity of the skills acquired and the differences in educational pathways that can influence learning outcomes, and the transversal skills developed. These programs are all undergraduate (bachelor-level) degrees offered at a single Italian university. The inclusion of students from the first, second, and third year of study allows for consideration of different stages in the academic path, which may influence the development and manifestation of transversal skills. Although these students had already selected a specific degree program, their level of familiarity with advanced disciplinary content varies depending on their year of enrollment. This distinction is particularly relevant to understanding the applicability of the serious game across different levels of academic maturity. Admission to these programs is based on general merit criteria (e.g., final grades from secondary school) and does not require formal aptitude testing or structured evaluation of soft skills. Therefore, while students may have selected their degree program based on interest or perceived compatibility, it cannot be assumed that this choice was guided by an in-depth assessment of their transversal competencies. These aspects introduce considerations about the timing and context in which the study was conducted. While the findings offer insight into the relationship between transversal skills and academic pathways, it is acknowledged that the most effective application of such predictive models may occur earlier in a student's educational trajectory, as will be further discussed in the limitations and future directions section.

Table 1
Descriptive statistics of serious Game participants by subgroups.

Subgroups		No. of Respondents	Percentage
Groups	Smartphone	109	51.7
	Desktop	102	48.3
Age	18–20	109	51.7
	21–25	92	43.6
	≥26	10	4.7
Gender	Male	87	41.2
	Female	124	58.8
Year of Study	1	83	39.3
	2	69	32.7
	3	59	28.0
Daily Screentime (hrs)	[1–2]	18	8.5
	]2–3]	1	0.5
	]3–4]	102	48.3
	]4–5]	3	1.4
	]5–6]	68	32.2
	]6–7]	2	0.9
	>7	17	8.1
Course	EBA	87	41.2
	SMAS	86	40.8
	ML	38	18.0

The "Subgroups" column specifies the categories into which the participants were divided, such as the type of device used (smartphone or desktop), age, gender, year of study, daily screen time, and degree program. The "Number of Respondents" column indicates the total number of participants for each category. The "Percentage" column shows the percentage of participants for each category relative to the total sample.

The study utilized a serious game to assess transversal skills such as time management, priority management, and planning. The game was accessible via both desktop PCs and smartphones, allowing a comparison between these two platforms. In addition to the game, various questionnaires were employed, including the COTI-33, composed of 33 items and validated by Agranovich et al. (2021), to evaluate participants' time perception.

The study involved 211 participants, recruited from various courses at a single university in Italy, divided into two groups: one using smartphones and one using desktop PCs. Data collection for the technological groups included a pre-test questionnaire to gather demographic information and previous exposure to serious games, the use of the serious game to assess transversal skills, and a post-test questionnaire to evaluate the perception of the game. The data collection process began with an explanation of the study's purpose to the participants, followed by the signing of the informed consent form. Only those who signed the consent form had access to the online serious game. During the game, which lasted about 30 min, participants had to manage various tasks such as meetings, emails, and phone calls in an organizational context, evaluated based on their time management, priority management, and planning skills. The serious game, designed to offer flexibility, eliminated the need for manual evaluation, with the online system providing final scores based on the participants' inputs. This mode allowed participants to complete the game at their convenience, without urgency. The serious game is set in an organizational context, where participants must manage various managerial tasks, such as personal meetings, emails, and phone calls, before going on vacation. This design follows the established in-basket exercise methodology, a widely validated assessment technique in organizational psychology where participants assume a managerial role and process accumulated workplace materials under time constraints (Schippmann et al., 1990). The in-basket format effectively simulates realistic decision-making scenarios and has been extensively used to evaluate soft skills such as prioritization, delegation, and time allocation in professional contexts. The different stages of the game are represented in Fig. 2.

The first step in the game involves accepting the privacy notice for data collection and selecting the character/gender. The second phase allows participants to get an overview of the different scenarios, such as people to meet, phone calls, and emails. In this phase, participants must manage 11 emails, 5 phone calls, and 4 personal meetings, with a maximum time of 30 min. Each scenario varies in terms of urgency and importance, and participants are evaluated based on their management decisions. For clarity, each type of task within the serious game was explicitly designed to assess a specific transversal skill through realistic decision-making scenarios. For example, email tasks required students to prioritize and respond to messages based on urgency and relevance, simulating common academic and administrative communications and targeting planning abilities. Phone call tasks introduced unscheduled interruptions that tested time management skills by forcing students to reorganize their priorities dynamically. Meeting tasks involved evaluating the necessity and scheduling of different appointments, thus focusing on priority management under time constraints. These tasks were not generic but developed with defined educational objectives, aiming to enhance decision-making, organizational planning, and self-regulation in contexts analogous to real academic and professional settings.

Once all scenarios are completed, the final assessment of skills is provided in real-time, measuring time management, priority management, and planning. After completing the assessment, participants in the technological groups filled out a post-test questionnaire to express their perceptions of using serious games in the educational context.

Finally, the participants' academic records were consulted to obtain their Grade Point Average (GPA), used as an indicator of academic performance.

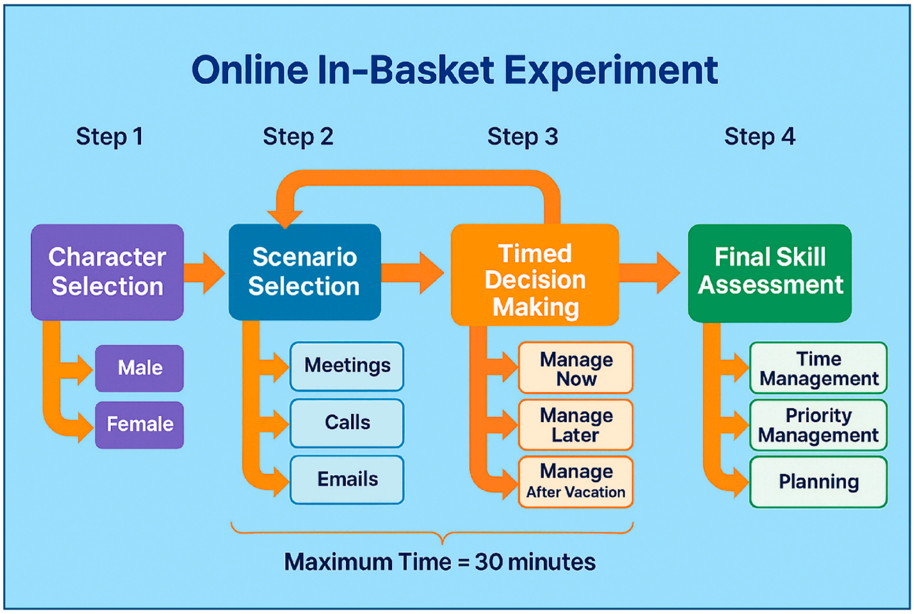

Fig. 2. The different stages of the online serious game for the assessment of transversal skills used in this study.

Since the overall data collection process is divided into three parts for the technological group (i.e., pre-test questionnaire, serious game for transversal skills assessment, and post-test questionnaire), the discussion regarding the different measures will be highlighted similarly. Below are the details of the three main steps and the subsequent measures from which the data are obtained:

- **Pre-test Questionnaire:** The pre-test questionnaire collected demographic information (age, gender) and academic indicators (year of study, course, GPA), as well as previous exposure to serious games. The questionnaire items were reviewed by experts to ensure content validity, and a pilot test showed high internal consistency with a Cronbach's alpha of 0.85.
- **Serious Game for Transversal Skills Assessment**: The serious game assessed time management, planning, and priority management through a storyline and professional context. The game scenarios were designed based on real managerial tasks, enhancing ecological validity. Test-retest reliability showed a correlation coefficient of 0.9 between two separate game sessions. A preliminary investigation of the online serious game used in this study is discussed in VS, F., Marengo, Pagano, Pange, and Piccinno (2022); Pagano and Marengo (2021); Madleňák, D'Alessandro, Marengo, Pange, and Ivanneszmélyi (2021).
- **Post-test Questionnaire:** The post-test questionnaire assessed technological factors, comfort and usage levels, and time perception using the COTI-33 questionnaire. This tool has been validated in previous studies and showed excellent internal consistency with a Cronbach's alpha of 0.92. The post-test allowed for examining the effect of technology on overall engagement with the serious game and on the scores obtained for different transversal skills. If participants delay urgent and high-priority activities, their overall priority management score will decrease. Conversely, if participants manage low-priority and non-urgent tasks urgently, this will also negatively reflect on the final priority management assessment. Participants' ability to effectively assess the importance of various tasks leads to a higher priority management score. Similarly, participants' ability to effectively plan different tasks in light of the provided contextual information and available business resources is reflected in their planning management skills.

The responses to the various questions administered in the study were collected using a Likert scale. This scale has five levels of agreement or disagreement, each associated with a specific score.

3.3. Model selection and development

Participants could respond to questions with "Strongly Agree," corresponding to a score of 2, "Agree," with a score of 1, "Neutral," with a score of 0, "Disagree," with a score of −1, and "Strongly Disagree," with a score of −2. This system allowed for quantifying participants' opinions and perceptions in a structured and analyzable manner. University exam scores are expressed out of thirty. All statistics related to transversal skills and GPA are reported in Table 2.

"N" represents the number of participants for each measure. "Mean" shows the average score, while "Median" indicates the central value of the scores. "SD" (Standard Deviation) measures the dispersion of scores around the mean. The columns "Minimum" and "Maximum" report the lowest and highest scores obtained. The data show that the average scores for priority management, planning, and time management are relatively high, indicating good competence in these areas. The GPA scores, which range from 18 to 30 with an average of 26.12, reflect the academic performance of the participants.

In the data preprocessing phase (orange box in Fig. 1) within the machine learning environment, we performed the following operations:

- Initial Dataset: The initial dataset comprised 211 instances.
- Feature Selection: We selected the 13 features under study using the "select column" widget. The selected features include 4 numeric variables (Priority, Planning, Time, and GPA) and 6 categorical variables (5 subgroups and 1 target variable), presented in Tables 1 and 2 This selection ensures that all relevant variables are included in the analysis.

Table 2
Predictive model features related to students' soft skills and academic performance (GPA): descriptive statistics.

	N	Mean	Median	SD	Minimum	Maximum
Priority	211	2.80	3.00	0.664	0.000	4.25
Planning	211	3.70	4.00	0.784	1.000	6.00
Time	211	2.74	2.80	1.000	0.600	4.60
GPA	211	26.12	27.00	2.757	18.000	30.00

- Standardization of Variables: We standardized the selected variables using the "continuize" widget, setting the mean to 0 and the standard deviation to 1. Standardization is crucial to ensure that the variables are comparable and the analysis is accurate. The general formula for standardizing a dataset with n variables is as follows:

$$Z_i = \frac{(X_i - \mu_i)}{\sigma_i} \; for \; i = 1, 2, \ldots, n. \tag{1}$$

where:

Z_i is the standardized value of the i-th variable.
X_i is the respective non-standardized value of the i-th variable.
μ_i is the mean of all observations in the i-th variable.
σ_i is the standard deviation of all observations in the i-th variable.
n is the number of variables (=4).

- Handling Outliers: We considered only inliers using the "outliers" widget. The parameters used were: Method = Local Outlier Factor; Contamination = 10 %; Neighbors = 20; Metric = Euclidean. This step excluded anomalous instances that could distort the analysis results. The number of inliers was 193, while the outliers were 18.

In the sampling phase (blue box in Fig. 1), we split the dataset into testing (20 %) and training (80 %) sets using the "Data Sampler" widget. The parameters used were: Fixed proportion of data = 80 %; Options: Replicable (deterministic) sampling; Stratify sample (when possible). This step is essential before the logistic regression phase to evaluate the model's performance on unseen data during training, ensuring the validity and generalizability of the results.

In this study, we employed logistic regression within a machine learning environment to develop a predictive model aimed at classifying students into three categories of courses of study: EBA, SMAS, and ML. Logistic regression is particularly suited for this task as it effectively models the probability of a categorical dependent variable based on multiple predictor variables, making it ideal for predicting categorical outcomes such as course enrolment based on participants' soft skills and GPA. The application of logistic regression in this context is justified by its ability to handle multiple predictors and provide probabilistic interpretations of the outcomes, which are crucial for understanding the factors influencing students' academic paths.

The logistic regression model was implemented using the Orange Data Mining software, with specific parameters set to optimize its performance. Regularization was applied using Ridge (L2) with a penalty parameter C set to 1, ensuring that the model does not overfit the training data. The class weights were not adjusted, maintaining a balanced approach to the classification task. After preprocessing and outlier removal, a total of 193 valid (inlier) instances were retained. From this dataset, 155 instances (80 %) were used for training and 42 instances (20 %) for testing. The model was built using four key features: Priority, Planning, Time, and GPA. The target variable for this analysis was the course of study.

The mathematical foundation of logistic regression in this case can be expressed as follows:

$$P(1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 * Priority + \beta_2 * Planning + \beta_3 * Time + \beta_4 * GPA)}} \tag{2}$$

Here, P(1|x) represents the probability of a student being classified into a specific course of study. The terms $\beta_0, \beta_1, \beta_2, \beta_3$ and β_4 denote the intercept and coefficients for the predictors Priority, Planning, Time, and GPA, respectively. This formula encapsulates the relationship between the predictor variables and the likelihood of the categorical outcome, providing a robust framework for predicting academic paths.

3.4. Predictive analysis and validation

Internal validation of the model was performed through stratified 10-fold cross-validation. This method involves partitioning the data into ten subsets, training the model on nine subsets, and testing it on the remaining one. This process is repeated ten times, with each subset serving as a test set once. Cross-validation is a rigorous method for assessing the model's generalizability and is crucial for preventing overfitting, thereby ensuring that the model performs well on new, unseen data.

The subsequent section, "Results and Discussions", will delve into the choice of the predictive model and its performance metrics. This section will critically evaluate the model's effectiveness using various metrics such as AUC-ROC, accuracy, precision, recall, F1 score, and MCC. These metrics will provide a comprehensive assessment of the model's ability to predict the most suitable course of study based on participants' soft skills and academic performance, ultimately contributing to a better understanding of the factors that influence academic success and the potential of machine learning applications in educational settings.

In our study, we developed a predictive model to classify the pupils participating in the serious game into three categories of courses of study: EBA, SMAS, and ML, based on the participants' soft skills and GPA. The model's performance is evaluated using several crucial metrics to ensure accurate and reliable predictions:

AUC-ROC (Area Under the Curve - Receiver Operating Characteristics): This metric assesses the discriminative ability of the model across the outcome classes, either individually or in an aggregated manner through macro or micro averaging. A value of AUC-ROC equal to 1 indicates a perfect classification, while a value of 0.5 indicates a random classification.

$$AUC_{macro} = \frac{1}{n}\sum_{i=1}^{n} AUC_i \tag{3}$$

where n is the total number of classes and AUC_i is the AUC calculated for class i. For a specific class i in a multiclass context, where class i is considered positive and all other classes are combined as negative, the AUC, denoted as AUC_i, is calculated through the integral:

$$AUC_i = \int_0^1 TPR_i \left[FPR_i^{-1}(u)\right] du \tag{4}$$

Here, TPR_i and FPR_i are, respectively, the true positive rate and false positive rate for class i, calculated across varying decision thresholds u. This integral covers all possible decision thresholds, providing a comprehensive measure of predictive accuracy for class i.

Accuracy: Accuracy measures the percentage of correctly classified cases relative to the total number of cases. It's a general metric that provides an overview of the overall effectiveness of the model.

$$Accuracy = \frac{\sum_{i=1}^{n} True\ Positives_i + \sum_{i=1}^{n} True\ Negatives_i}{\sum_{i=1}^{n} Total\ Population_i} \tag{5}$$

Precision: Precision measures the proportion of true positive cases correctly identified relative to all cases identified as positive by the model. It quantifies the accuracy of positive predictions made by the model.

$$Precision_i = \frac{TruePositives_i}{True\ Positives_i + \sum_{j=1, j\neq i}^{n} False\ Positives_{ji}} \tag{6}$$

where $False\ Positives_{ji}$ is the number of times that class j was incorrectly predicted as class i.

Recall (Sensitivity): Recall measures the proportion of true positive cases correctly identified relative to all actually positive cases. It quantifies the model's ability to capture all positive cases, minimizing false negatives.

$$Recall_i = \frac{TruePositives_i}{True\ Positives_i + \sum_{j=1, j\neq i}^{n} False\ Negatives_{ij}} \tag{7}$$

where $False\ Negatives_{ij}$ is the number of times that class ii was incorrectly not predicted as class i.

F1 Score: The F1 score is a harmonic mean of precision and recall. This metric is useful when a balance between precision and recall is desired, and a single measure is sought to evaluate the model's performance.

$$F1\ Score_i = 2 \times \frac{Precision_i\ x\ Recall_i}{Precision_i + Recall_i} \tag{8}$$

The overall F1 score can be calculated by averaging the F1 scores for each class.

Matthews Correlation Coefficients (MCC): MCC is a measure of the overall quality of the model's predictions, considering both positive and negative cases. It ranges from −1 to 1, where 1 indicates perfect prediction, 0 indicates random prediction, and −1 indicates completely wrong prediction.

$$MCC = \frac{CxS - \sum_k P_k x\ T_k}{\sqrt{\left(S^2 - \sum_k P_k^2\right) \times \left(S^2 - \sum_k T_k^2\right)}} \tag{9}$$

where:

C is the sum of all elements on the diagonal of the confusion matrix (correct predictions).

S is the total sum of all elements in the confusion matrix.

P_k is the sum of all elements in column k of the confusion matrix (predictions for class k).

T_k is the sum of all elements in row k of the confusion matrix (actual values for class k).

These metrics are calculated using the confusion matrix (Fig. 4), which is structured with columns representing the predicted classes and rows representing the actual classes. "True Positives" for a class i are those elements where both the predicted and the actual class are i. Similarly, "True Negatives" for class i are all the correct predictions that do not belong to class i, "False Positives" are those instances where a class other than ii is incorrectly predicted as class i, and "False Negatives" are those instances where class i is incorrectly predicted as not being class i.

3.5. Data Envelopment Analysis (DEA)

To further evaluate the effectiveness of serious games in assessing soft skills, we conducted a Data Envelopment Analysis (DEA) on the 193 participants in the Serious Game, obtained after excluding 18 outliers from the initial dataset of 211 students. These outliers were identified during the data preprocessing phase using the Local Outlier Factor (LOF) method with a 10 % contamination threshold, ensuring that the analysis focused on reliable and representative data. DEA is a non-parametric method used to assess the relative efficiency of Decision-Making Units (DMUs) that convert multiple inputs into multiple outputs. This approach was chosen for various methodological and practical reasons:

- Assessment of Relative Efficiency: DEA allows us to compare the relative efficiency of participants in the demographic and behavioral subgroups, using soft skills as inputs and academic performance (GPA) as the output. This approach is particularly useful for identifying whether one technology is more effective in optimizing the use of soft skills to improve academic performance.
- Output-Oriented Model with Variable Returns to Scale (VRS): We chose an output-oriented model to focus on optimizing the result (GPA) relative to the available inputs (soft skills). Variable returns to scale (VRS) were adopted to reflect the reality that DMUs (in this case, students) can operate at different scales, recognizing that not all students operate under the same conditions or with the same resources.
- Non-Parametric Approach: DEA is particularly useful in educational contexts where inputs and outputs may not follow normal distributions and where relationships between variables are not necessarily linear. This makes DEA an appropriate method for our study, where soft skills and academic performance can vary widely among individuals.

We collected data on soft skills (time management, priority management, planning) and the academic performance (GPA) of the participants. After standardizing the three input features and the output feature using the (1) formula, we positivized all standardized features by applying the following formula:

$$X_{POSITIVE} = Z_i + \left|\min\left(Z_{\text{Planning}}, Z_{\text{Priority}}, Z_{\text{Time}}, Z_{\text{GPA}}\right)\right| + 1, for\ i = 1, 2, \ldots, n. \tag{10}$$

where $\left|\min\left(Z_{\text{Planning Management}}, Z_{\text{Priority Management}}, Z_{\text{Time Management}}, Z_{\text{GPA}}\right)\right|$ is the minimum value among all four standardized features (three inputs and one output) plus 1. Positivization can help eliminate any negative effects of variables with negative values, which could negatively impact efficiency evaluations.

We then applied the following DEA formula to the positivized Input/Output variables in an output-oriented model with variable returns to scale (VRS):

$$Max\ \text{s}, \text{subject to} : s \le \frac{X_{POSITIVE\ INPUT_j}}{X_{POSITIVE\ OUTPUT_j}}, for\ j = 1, \ldots, m \tag{11}$$

where $X_{POSITIVE\ INPUT_j}$ and $X_{POSITIVE\ OUTPUT_j}$ are the values of input and output for j-th DMU and m is the total number of decision-making units (=193).

4. Results

In this section, we present the empirical findings of our study, focusing on the performance metrics of the logistic regression model and other predictive algorithms in classifying students into their respective degree programs based on soft skills and GPA. We report the statistical outcomes of model comparisons, feature importance analysis through SHAP values, demographic analysis using Kruskal-Wallis ANOVA, and Data Envelopment Analysis efficiency scores across the identified subgroups.

4.1. Performance and choice of the predictive model

The main metric chosen for the evaluation of predictive algorithms is the AUC (Area Under the Curve), as it provides a comprehensive measure of the model's ability to discriminate between different classes. Other metrics such as Classification Accuracy (CA), F1 Score, Precision (Prec), Recall, and Matthews Correlation Coefficient (MCC) were also considered. To ensure a balanced and robust evaluation, Stratified 10-fold Cross Validation was used for sampling.

After comparing various algorithms, including Logistic Regression, Neural Network, SVM, Random Forest, Gradient Boosting, Naive Bayes, AdaBoost, and Stochastic Gradient Descent, it was found that Logistic Regression demonstrated solid and balanced performance across all key metrics. Logistic Regression achieved the highest AUC (0.814 $\pm$ 0.032) among the tested models, with all other algorithms showing significantly lower performance ($p < 0.05$, Nemenyi post-hoc test). Additionally, Logistic Regression maintained a balance between precision (0.714 $\pm$ 0.035) and recall (0.703 $\pm$ 0.041), with a good F1 Score (0.683 $\pm$ 0.038) and a robust MCC (0.504 $\pm$ 0.056), showcasing its reliability in predicting the most suitable course of study based on participants' soft skills and GPA.

These results support the selection of Logistic Regression as the preferred predictive model for our study. The combination of high performance and balanced metrics makes Logistic Regression an ideal choice for providing reliable insights into the most suitable course of study for each participant. To ensure statistical rigor in model comparison, we employed the Friedman test, a non-parametric statistical test specifically designed for comparing multiple algorithms across multiple performance metrics on the same dataset.

The Friedman test ranks algorithms for each metric and evaluates whether the observed differences in rankings are statistically significant, without assuming normal distribution of the data. The test yielded a significant result ($\chi^2 = 47.23$, df = 7, $p < 0.001$), indicating that the performance differences among the eight algorithms are not due to random variation. Subsequently, Nemenyi post-hoc analysis was conducted to identify specific pairwise differences between algorithms. This analysis revealed that Logistic Regression significantly outperformed all competing algorithms across all six evaluation metrics ($p < 0.05$ for all pairwise comparisons), providing robust statistical evidence for its selection as the optimal predictive model. Below, Fig. 3 illustrates the performance metrics for all evaluated algorithms, clearly highlighting the superiority of Logistic Regression in terms of AUC, CA, F1 Score, Precision, Recall, and MCC.

The confusion matrix for Logistic Regression, showing the proportion of predicted values, provides additional insight into the model's classification performance. The matrix (Fig. 4) is detailed as follows:

Fig. 5 presents the ROC curves for all three target classes in a comparative layout, facilitating direct comparison of the model's discriminative performance across EBA, ML, and SMAS degree programs.

The ROC curves evaluate the performance of the classification model. The diagonal black line represents a random classifier. The red curve shows the actual performance of the model: the closer it is to the top left corner, the better the model's ability to discriminate between classes. The gray area (AUC) represents the model's accuracy, with an AUC close to 1 indicating high accuracy.

4.2. Importance of predictive variables on academic pathways

Our study revealed how different predictive variables significantly influence the model used to classify students based on their academic pathways. The Grade Point Average (GPA) emerged as the most determining factor, showing a profound impact on the model's performance. This result underscores the importance of GPA as a measure of academic performance, providing a clear indication of students' capabilities and commitment. It is important to clarify that the GPA used in our model refers to the students' partial academic performance at the time of the serious game, calculated on the basis of the exams already completed after enrollment in their respective degree programs. This GPA is not a final graduation score, and the model does not predict future course selections, but rather classifies students according to their observed academic and soft skill profiles. In fact, the removal of GPA from the model caused a drastic decrease in the Area Under the Curve (AUC) by 0.316, with a standard deviation of 0.012, demonstrating how crucial

		Predicted			
		EBA	ML	SMAS	Σ
Actual	EBA	82.5 %	0.0 %	17.4 %	67
	ML	0.0 %	66.7 %	20.9 %	22
	SMAS	17.5 %	33.3 %	61.6 %	66
	Σ	63	6	86	155

Fig. 4. Confusion matrix for logistic regression.

this indicator is for predicting different academic courses.

In contrast, transversal skills such as time management, planning, and priority management show a relatively minor impact. For instance, removing time management from the model caused a decrease in AUC by 0.023 with a standard deviation of 0.010, while planning and priority management caused decreases of 0.012 (std 0.004) and 0.010 (std 0.004), respectively. These results indicate that although transversal skills are useful, their predictive contribution is significantly lower compared to GPA.

Educational analytics from the final dataset of 193 inlier students revealed significant competency gaps requiring targeted pedagogical interventions. Priority management emerged as the most critical educational challenge across all programs, with only 4.9 % of EBA students, 2.5 % of SMAS students, and 10.7 % of ML students achieving high proficiency levels (>3.5 on 5-point scale). In contrast, planning management showed stronger development with 64.6 % of EBA, 57.5 % of SMAS, and 75.0 % of ML students reaching high proficiency. Academic performance distribution analysis revealed distinct program learning profiles: ML students demonstrated exceptional achievement with 82.1 % classified as high performers (GPA >27), SMAS students showed strong performance with 46.3 % high performers, while EBA students exhibited more diverse outcomes with 40.2 % classified as low performers (GPA <24) alongside 14.6 % high performers. Platform-specific analytics confirmed technology-neutral learning effectiveness, with desktop PC users (n = 93, 48.2 %) achieving average GPA of 26.01 and smartphone users (n = 100, 51.8 %) demonstrating average GPA of 26.22, showing no statistically significant difference ($p > 0.05$) and supporting inclusive educational technology deployment across diverse institutional contexts.

4.3. Predictive results with SHAP values

The results obtained from the predictive analysis using the logistic regression model demonstrated a strong ability to accurately classify participants into their respective degree programs based on their

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.814 ± 0.032	0.703 ± 0.041	0.683 ± 0.038	0.714 ± 0.035	0.703 ± 0.041	0.504 ± 0.056
Neural Network	0.794 ± 0.038*	0.677 ± 0.045*	0.667 ± 0.042*	0.676 ± 0.039*	0.677 ± 0.045*	0.461 ± 0.061*
Naive Bayes	0.781 ± 0.041*	0.665 ± 0.048*	0.668 ± 0.036*	0.674 ± 0.044*	0.665 ± 0.048*	0.454 ± 0.067*
SVM	0.775 ± 0.035*	0.671 ± 0.043*	0.640 ± 0.051*	0.743 ± 0.041*	0.671 ± 0.043*	0.456 ± 0.059*
Random Forest	0.760 ± 0.047*	0.632 ± 0.052*	0.626 ± 0.048*	0.626 ± 0.055*	0.632 ± 0.052*	0.388 ± 0.071*
Gradient Boosting	0.743 ± 0.051*	0.600 ± 0.056*	0.594 ± 0.053*	0.589 ± 0.048*	0.600 ± 0.056*	0.337 ± 0.078*
SGD	0.714 ± 0.063*	0.690 ± 0.049*	0.641 ± 0.057*	0.605 ± 0.062*	0.690 ± 0.049*	0.479 ± 0.068*
AdaBoost	0.606 ± 0.072*	0.516 ± 0.071*	0.512 ± 0.069*	0.512 ± 0.074*	0.516 ± 0.071*	0.212 ± 0.089*

*p < 0.05 compared to Logistic Regression (Nemenyi post-hoc test)

Fig. 3. Performance Metrics for Predictive Algorithms (mean ± standard deviation across 10-fold cross-validation; *$p < 0.05$ vs. Logistic Regression).

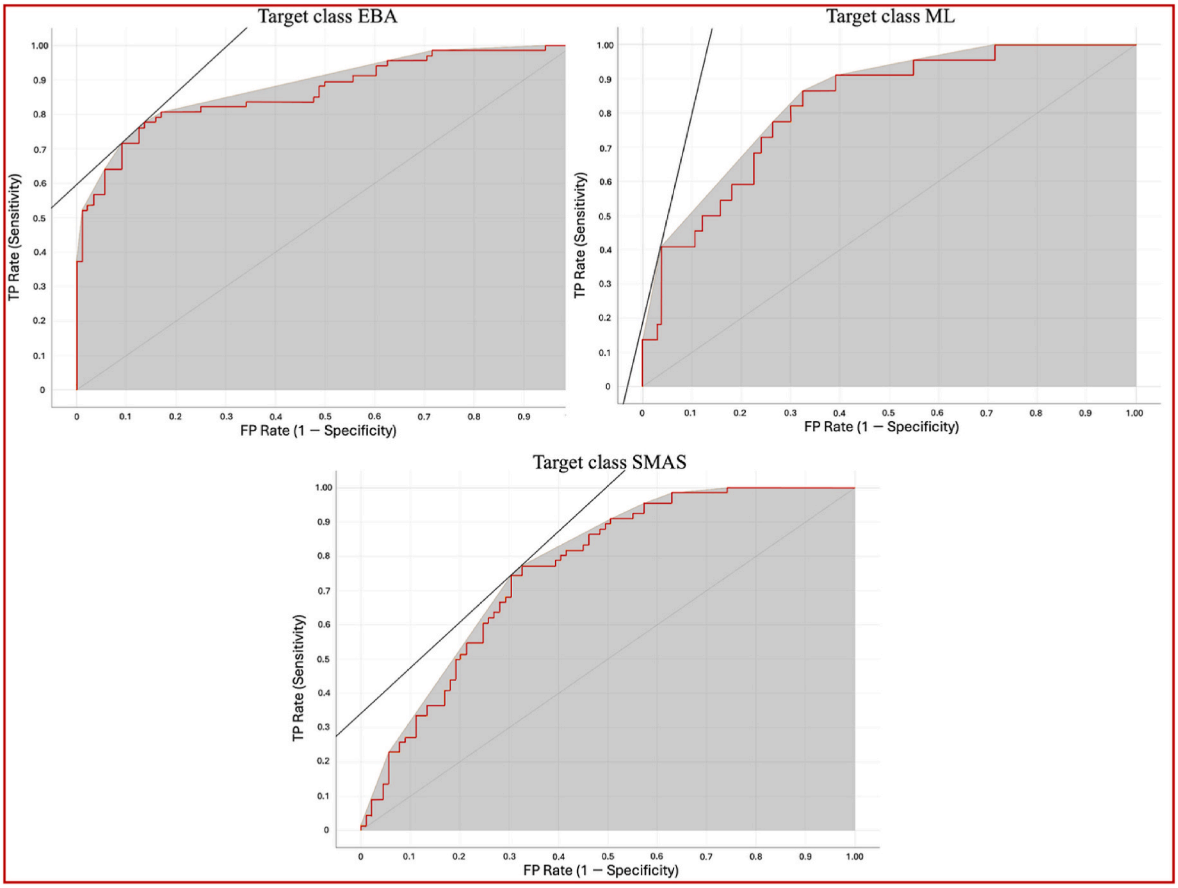

Fig. 5. ROC curves for Economic business administration (EBA), modern literature (ML), and Sports and Motor activity sciences (SMAS).

transversal skills and academic performance (GPA). Fig. 7 illustrates the SHAP value bar plots for the three target classes: Economic Business Administration (EBA), Sports and Motor Activities (SMAS), and Modern Literature (ML). These plots provide a visual and quantitative representation of the influence of predictive variables, offering a detailed understanding of the dynamics underlying the model's predictions. Red bars indicate a positive impact on the prediction, pushing the probability from the model's base value to the predicted value, as represented by the gray boxes. Blue bars indicate a negative impact on the prediction, pushing the probability downward. The white values within the bars are the SHAP values that represent the magnitude of each variable's impact.

The bar plot for the EBA class highlights that GPA has a significant negative impact (−0.387035), suggesting that higher GPA values decrease the probability of being classified as EBA, while lower GPA values increase this probability relative to the model's baseline. Planning shows a moderate positive impact (0.0298767), indicating that good planning skills increase the probability of classification in EBA. Time management (0.00882427) and priority (−0.0169707) have minor but relevant impacts. These results suggest that EBA classification is inversely associated with GPA, indicating that students with lower academic performance but strong planning and time management skills may be more likely to be classified in this program. This suggests that EBA may value practical competencies alongside or even over pure academic achievement.

For the SMAS class, GPA emerges as the most influential variable with a significant positive impact (0.122534). However, both planning (−0.0386874) and time management (−0.0633577) show significant negative impacts, suggesting that better skills in these areas could paradoxically reduce the probability of being classified as SMAS. Priority has a slightly negative impact (−0.00176024). These results indicate an interesting dynamic where high academic performance increases the probability of being classified into the SMAS program, while excellent transversal skills in planning and time management may not be as advantageous for this specific classification.

In the bar plot for the ML class, GPA has a significant positive impact (0.264501), confirming that a higher GPA increases the probability of being classified as ML. Time management (0.0545334) also has a moderate positive impact, highlighting the importance of this skill in predicting success in ML. Planning (0.00881073) and priority (0.018731) have minor but positive impacts. This suggests that, in addition to academic performance, transversal skills like time management and planning are important factors contributing to the classification into the Modern Literature.

4.4. Statistical analysis of demographic and behavioral variables using Kruskal-Wallis ANOVA

As the next step, we conducted a Kruskal-Wallis ANOVA analysis, using the predicted values from the logistic regression as the grouping variable and Age, Gender, Year of Study, Technology Groups, and Daily Screentime (hours) as the dependent variables. In order to better contextualize the predictions made by the logistic regression model, we conducted a Kruskal-Wallis ANOVA to test whether the three predicted student groups (EBA, SMAS, ML) significantly differ in terms of their demographic and behavioral profiles. Specifically, this analysis investigates whether variables such as Age, Gender, Year of Study, Technology Group, and Daily Screentime show different distributions across the predicted classes. The purpose of this analysis is not to improve the classification performance, but rather to explore whether the model-based groupings also reflect meaningful differences in background characteristics. Identifying such patterns allows us to interpret the model's predictions in a more nuanced way, supporting the design of personalized orientation strategies and educational interventions tailored to the specific needs of students in each academic area. All the variables considered are categorical. The Kruskal-Wallis ANOVA analysis was chosen to determine if there are significant differences between the groups predicted by the logistic regression model with respect to the demographic and behavioral variables. This type of non-parametric analysis is particularly useful for categorical variables and when the assumptions of normality are not met. The Kruskal-Wallis analysis was chosen because it does not require the data to follow a normal distribution, making it suitable for categorical variables. Additionally, it is less sensitive to outliers and skewed distributions compared to parametric tests and is particularly useful when the dependent variables are ordinal or categorical.

The analysis results showed significant differences between the predicted groups concerning the distribution of age groups, with a χ^2 value of 25.28, df = 2, and $p < 0.001$. No significant differences were observed in the distribution of gender between the different predicted groups, with a χ^2 value of 3.98, df = 2, and p = 0.136, indicating that gender does not play a determining role in academic orientation and course choice according to the predictions of the logistic regression model. The analysis revealed significant differences between the predicted groups in terms of the year of study, with a χ^2 value of 58.54, df = 2, and $p < 0.001$. This could reflect how academic experience influences transversal skills and academic performance (Table 3). No significant differences were found between smartphone and desktop users in the different predicted groups, with a χ^2 value of 1.80, df = 2, and p = 0.406, suggesting that the type of device used for serious games does not significantly impact the development of skills and academic performance predicted by the model. Finally, the analysis showed no significant differences in daily screentime between the predicted groups, with a χ^2 value of 3.56, df = 2, and p = 0.168 (Table 3). For variables that showed significant differences, we conducted Dwass-Steel-Critchlow-Fligner pairwise comparisons to identify which groups these differences exist. The results of the pairwise comparisons for Age and Year of Study are presented below.

The results of the pairwise comparisons for the variable Age indicate that there are significant differences between EBA and ML (W = 5.50, $p < 0.001$) and between EBA and SMAS (W = 6.24, $p < 0.001$), while there are no significant differences between ML and SMAS (W = −2.10, p = 0.297) (Table 4). For the Year of Study, there are significant differences between EBA and ML (W = 6.34, $p < 0.001$) and between EBA and SMAS (W = 10.32, $p < 0.001$), while there are no significant differences between ML and SMAS (W = −2.49, p = 0.183) (Table 5).

4.5. *Data Envelopment Analysis: results and implications*

The efficiency scores s for each DMU were then calculated. A score of 1 indicates efficiency, while scores below 1 indicate relative inefficiency. Descriptive statistics for the identified efficiency scores are presented in Table 6. In this study, the efficiency score represents the relative capability of each student to transform their soft skills—namely, time management, planning, and priority management—into academic performance, as measured by GPA. Each student was modeled as a Decision-Making Unit (DMU), with the three soft skills serving as input variables and the GPA as the output. The DEA output-oriented model with variable returns to scale (VRS) enabled us to assess how efficiently students use their soft skill resources to maximize academic performance. A score of 1 denotes full efficiency, indicating that the student is performing on the frontier, while a score below 1 reflects relative inefficiency in converting soft skills into academic results. This interpretation provides a valuable insight into the relationship between behavioral competencies and academic outcomes, supporting more personalized and targeted educational interventions.

Table 3
Results of the Kruskal-Wallis ANOVA for the variables Age, Gender, Year of Study, Technology Groups, and Daily Screentime (hours).

	χ^2	df	p
Age	25.28	2	<0.001
Gender	3.98	2	0.136
Year	**58.54**	**2**	**< .001**
Technology Groups	1.80	2	0.406
Daily Screentime (hours)	3.56	2	0.168

Table 4
Dwass-Steel-Critchlow-Fligner pairwise comparisons for the variable Age.

		W	p
EBA	ML	5.50	<0.001
EBA	SMAS	6.24	<0.001
ML	SMAS	−2.10	0.297

Table 5
Dwass-Steel-Critchlow-Fligner pairwise comparisons for the variable Year of Study.

		W	p
EBA	ML	6.34	<0.001
EBA	SMAS	10.32	<0.001
ML	SMAS	−2.49	0.183

The Shapiro-Wilk test for normality indicated that the efficiency scores were not normally distributed (W = 0.921, $p < 0.001$). This non-normal distribution necessitated the use of a non-parametric statistical test to compare the efficiency scores between demographic and behavioral subgroups. We selected the Kruskal-Wallis ANOVA, a non-parametric method suitable for comparing more than two groups when the assumption of normality is not met.

This approach was implemented in a machine learning environment, following what was outlined by the PDA methodology applied in the study of hospital facility efficiency (Santamato et al., 2024).

The results of the Kruskal-Wallis analyses indicate that there are significant differences in DEA efficiency scores among different courses ($\chi^2 = 80.3$, df = 2, $p < 0.001$) (Table 7), different age groups ($\chi^2 = 29.0$, df = 2, $p < 0.001$) (Table 9), and between genders ($\chi^2 = 5.51$, df = 1, p = 0.019) (Table 11). The median efficiency score is 0.797 for females and 0.766 for males. However, there are no significant differences in DEA scores among different technology groups ($\chi^2 = 0.0843$, df = 1, p = 0.771) (Table 14) and daily screentime usage ($\chi^2 = 8.40$, df = 6, p = 0.210) (Table 15) (see Table 12).

The Dwass-Steel-Critchlow-Fligner pairwise comparisons show that, among courses, there are significant differences between EBA and ML (W = 9.47, $p < 0.001$), EBA and SMAS (W = 10.69, $p < 0.001$), and ML and SMAS (W = −4.61, p = 0.003) (Table 8). For age, there are significant differences between the 18–20 age group and the 21–25 age group (W = 7.36, $p < 0.001$), and a marginally significant difference between the 18–20 age group and the 26+ age group (W = 3.29, p = 0.052), but no significant difference between the 21–25 age group and the 26+ age group (W = −1.10, p = 0.719) (Table 10). For year of study, there are significant differences between the first and second years (W = 9.62, $p < 0.001$), the first and third years (W = 10.49, $p < 0.001$), but not between the second and third years (W = 2.70, p = 0.137) (Table 13).

These results suggest that the efficiency of participants in the Serious Game varies significantly depending on the course of study, age, and gender. These differences can be used to customize educational interventions and improve student orientation and support, taking into account their demographic and behavioral characteristics. The scientific implications of these results are manifold. Firstly, they confirm the importance of considering demographic and behavioral variables when evaluating the effectiveness of serious games. Secondly, they suggest

Table 6
Descriptive statistics for DEA efficiency scores.

								Shapiro-Wilk	
	N	Missing	Mean	Median	SD	Minimum	Maximum	W[a]	p
DEA Score	193	0	0.740	0.797	0.187	0.186	1.00	0.921	<0.001

[a] "W" in the Shapiro–Wilk test is a test statistic that assesses the normality of data distribution. Values closer to 1 indicate a greater conformity of the data to a normal distribution.

Table 7
Results of Kruskal-Wallis ANOVA for DEA SCORES in courses.

	χ^2	df	p
DEA_SCORES	80.3	2	<0.001

Table 8
Dwass-Steel-Critchlow-Fligner pairwise comparisons for DEA SCORES in Courses.

		W	p
EBA	ML	9.47	<0.001
EBA	SMAS	10.69	<0.001
ML	SMAS	−4.61	0.003

Table 9
Results of Kruskal-Wallis ANOVA for DEA SCORES by age.

	χ^2	df	p
DEA_SCORES	29.0	2	<0.001

Table 10
Dwass-Steel-Critchlow-Fligner pairwise comparisons for DEA SCORES by Age.

		W	p
18-20	21–25	7.36	<0.001
18-20	26++	3.29	0.052
21-25	26++	−1.10	0.719

Table 11
Results of Kruskal-Wallis ANOVA for DEA SCORES by gender.

	χ^2	df	p
DEA_SCORES	5.51	1	0.019

Table 12
Results of Kruskal-Wallis ANOVA for DEA SCORES by year of study.

	χ^2	df	p
DEA_SCORES	5.51	1	0.019

Table 13
Dwass-Steel-Critchlow-Fligner pairwise comparisons for DEA SCORES by Year of Study.

		W	p
1	2	9.62	<0.001
1	3	10.49	<0.001
2	3	2.70	0.137

that educational strategies should be adapted to meet the diverse needs of students, thereby maximizing the effectiveness of educational interventions and improving overall academic performance.

Table 14
Results of Kruskal-Wallis ANOVA for DEA SCORES by technology group.

	χ^2	df	p
DEA_SCORES	0.0843	1	0.771

Table 15
Results of Kruskal-Wallis ANOVA for DEA SCORES for time for daily use of screens.

	χ^2	df	p
DEA_SCORES	8.40	6	0.210

5. Discussion

In this section, we interpret and contextualize the empirical findings presented in the Results section. We discuss the implications of the logistic regression model's superior performance, analyze the meaning of predictive variable importance, examine the demographic patterns identified through statistical analysis, and explore the broader significance of our findings for educational guidance and serious games applications in higher education.

5.1. Interpretation of model performance

The confusion matrix indicates that the Logistic Regression model is particularly effective at correctly predicting EBA cases, with an accuracy of 82.5 % for actual EBA cases. However, there are some misclassifications, such as 17.5 % of SMAS cases being predicted as EBA and 33.3 % of SMAS cases being predicted as ML. This suggests that the model has some difficulty distinguishing between the SMAS and EBA classes, as well as between SMAS and ML. Additionally, ML cases are correctly classified 66.7 % of the time, while 20.9 % are erroneously predicted as SMAS. Overall, the model has demonstrated good discriminative ability, but there is room for improving the precision of predictions for the SMAS and ML classes.

The ROC curves for the three target classes (EBA, ML, SMAS) further illustrate the performance of the Logistic Regression model. Each ROC curve plots the True Positive Rate (Sensitivity) against the False Positive Rate (1-Specificity), providing a visual representation of the model's ability to distinguish between the classes. The curves show the model's high sensitivity and specificity for EBA classification, good performance in distinguishing ML cases, and solid ability to correctly identify SMAS cases.

The comparative ROC analysis demonstrates that the Logistic Regression model performs well across all target classes, with particularly strong performance for the EBA class. The unified presentation reveals areas for potential improvement, especially in enhancing the model's specificity for the ML and SMAS classes, while allowing for easy visual comparison of the AUC values and curve shapes across the three degree programs. In all three graphs, the red curve is significantly above the diagonal line, and the gray area is substantial, indicating that the model has a good ability to correctly distinguish between the SMAS, EBA, and ML classes.

5.2. Implications of predictive variables

Integrating academic performance with transversal skills could provide a more comprehensive picture of students' potential, improving predictions regarding their academic course choices and success probabilities. This combination of variables not only helps identify students' strengths and weaknesses but can also guide more targeted and personalized educational interventions, thus increasing the chances of success in an increasingly competitive academic environment.

Fig. 6 provides a clear visual representation of how GPA is the most influential variable in predicting academic courses, while transversal skills have a lesser impact. The analysis of SHAP values confirms that while GPA remains the primary parameter for predicting academic course allocation, transversal skills still play an important role in shaping students with a more complete and versatile skill set. This visual representation facilitates understanding the relative importance of each variable, offering a more detailed view of the dynamics influencing students' academic choices.

These educational analytics provide concrete evidence for implementing competency-based curriculum development and evidence-based pedagogical interventions. The identified priority management deficit across all programs (only 4.9–10.7 % achieving high proficiency) indicates systematic gaps in traditional educational approaches that emphasize theoretical knowledge over practical decision-making skills. This finding supports implementing AI-enhanced simulation-based learning modules that provide repeated practice in priority assessment scenarios, particularly important given the professional demands students will face in their respective fields. The stronger planning management skills (57.5–75.0 % high proficiency) suggest existing curricula effectively develop analytical and organizational thinking, providing a foundation for building more advanced soft skills competencies. The dramatic performance differences between programs (ML: 82.1 % high performers vs. EBA: 40.2 % low performers) validate the premise that different academic pathways require differentiated pedagogical approaches and support systems. EBA students' engagement with practical scenarios despite lower traditional academic metrics demonstrates that serious games can capture capabilities not reflected in x, supporting competency-based assessment and selection approaches that better align with professional skill requirements.

5.3. SHAP values interpretation

These findings provide important insights into the role of both academic performance and transversal skills in determining students' success in different academic fields. For EBA, the negative impact of GPA confirms that the field attracts students with lower traditional academic performance but strong practical skills and planning abilities, suggesting that EBA values applied competencies over purely academic achievements. In contrast, the SMAS results suggest a unique balance where high academic performance is essential, but overly strong planning and time management skills may not align with the practical and dynamic nature of the field. For ML, the positive impacts of both GPA and transversal skills underline the holistic requirements of this field, which values both intellectual capabilities and effective personal management.

The use of SHAP values in this analysis offers a transparent and interpretable method to understand the contributions of various predictors. This transparency is crucial for developing targeted educational interventions and personalized guidance strategies that can help students leverage their strengths and address their weaknesses effectively.

In conclusion, integrating SHAP values into predictive modeling provides a nuanced understanding of how different skills and performance metrics impact academic success across various fields. These insights can inform educators and policymakers in designing more effective curricula and support systems that cater to the diverse needs of students, ultimately enhancing educational outcomes and professional preparedness. The finding that lower GPA scores increase the likelihood of EBA classification is particularly intriguing and warrants further investigation. It may suggest that EBA programs attract students with a more practical orientation, whose strengths are not fully captured by traditional academic assessments. The soft skills measured by the serious game appear to be more predictive of success in business contexts than GPA alone. This challenges the assumption that grades are the most reliable selection criterion and points toward the potential value of a competency-based approach in business education. Future research should explore whether this trend reflects actual program needs or highlights areas for curriculum improvement.

5.4. Demographic patterns discussion

This suggests that age can influence the probability of success in specific degree programs (Table 3). Students of different age groups may have different approaches to studying and using educational technologies, which can reflect their academic performance and transversal skills. Students in more advanced years of study may have different skills and study habits compared to first-year students, thus influencing their probability of academic success. This result indicates that the time spent on digital devices does not significantly affect transversal skills and academic performance according to the predictions of the logistic regression model.

Post hoc analyses revealed that significant differences in the variables of age and year of study are mainly between the EBA class and the other two classes (ML and SMAS). This implies that students in Business Administration tend to have different demographic and behavioral characteristics compared to students in other classes. In particular, age seems to be a distinguishing factor for EBA students compared to those in ML and SMAS. EBA students might belong to more specific age groups, which could reflect a different phase of their academic career or particular educational needs. This difference can be used to develop

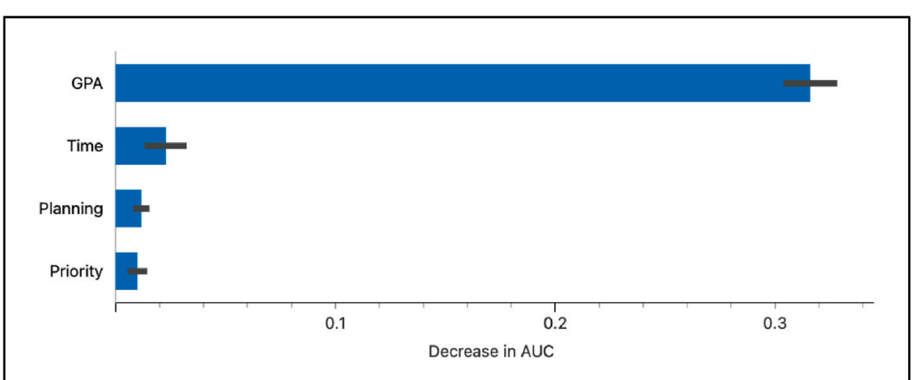

Fig. 6. Representation SHAP values quantifying feature impacts on model output.

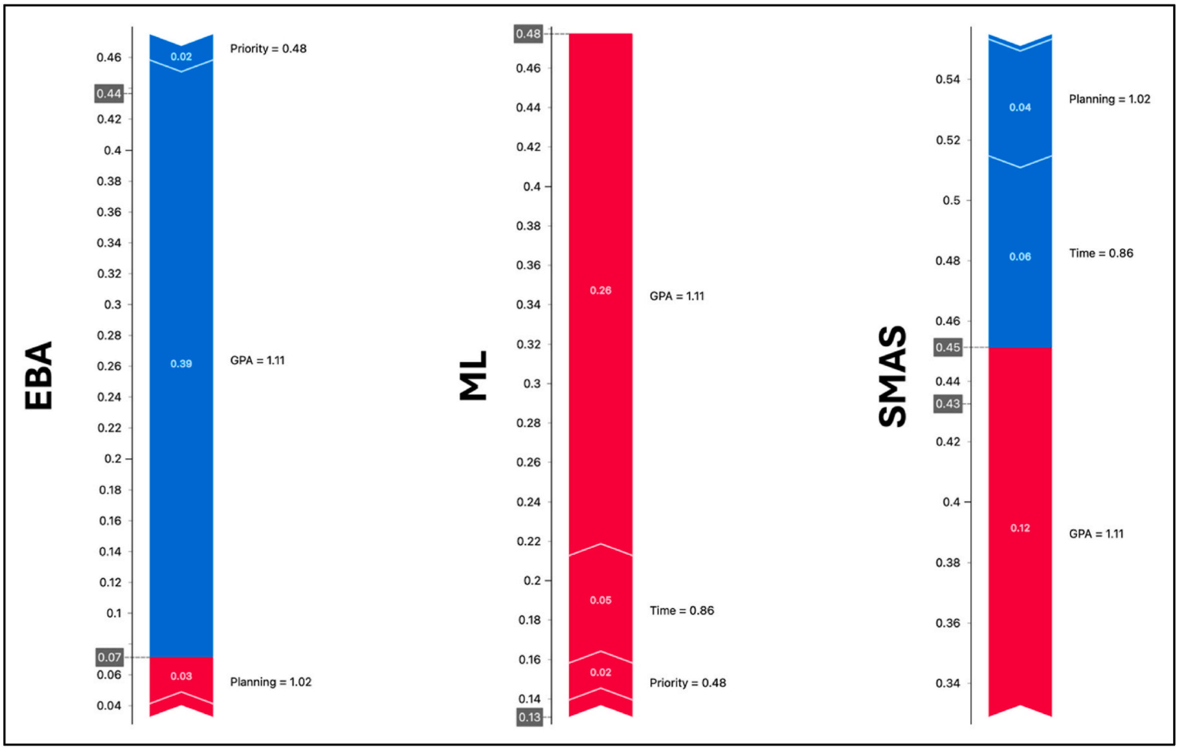

Fig. 7. SHAP value bar plots for the three target classes: Business Administration (EBA), Sports and Motor Activities (SMAS), and Modern Literature (ML).

more targeted orientation and support programs that take into account the specific age-related needs of EBA students. Regarding the year of study, significant differences between EBA and the other classes suggest that EBA students might have different academic experiences. Students in more advanced years of EBA might have developed more solid transversal skills or have greater familiarity with available educational resources compared to their peers in ML and SMAS. This may indicate the need for specific educational interventions to support students in the early years of EBA, to facilitate a smoother transition and improve their overall academic performance.

The results of the Kruskal-Wallis ANOVA provide further insights into the demographic and behavioral characteristics that influence academic success in various degree programs. The significant difference in the variables of age and year of study suggests that these factors should be considered when developing personalized educational interventions and orientation strategies. The variables of gender, technology groups, and daily screentime did not show significant differences, indicating that other factors might be more influential in determining academic success. The scientific implications of these results are manifold. Firstly, they confirm the importance of a holistic approach to academic orientation, which considers not only academic performance and transversal skills but also demographic characteristics and digital behaviors of students. Secondly, they suggest that educational technologies, such as serious games, should be designed considering the diverse needs and preferences of students to maximize the effectiveness of educational interventions. In summary, the Kruskal-Wallis ANOVA analysis offers a detailed view of the variables that influence academic success and provides a solid foundation for developing educational policies and personalized interventions aimed at improving student orientation and success.

5.5. DEA analysis implications

The use of Data Envelopment Analysis (DEA) to evaluate student efficiency in the context of Serious Games offers numerous methodological and practical advantages. Here are some key considerations:

- **Multidimensional Evaluation:** DEA allowed us to assess efficiency by simultaneously considering multiple inputs (soft skills) and one output (GPA). This holistic approach is particularly useful in complex educational contexts where performance cannot be measured by a single indicator.
- **Adaptability to Different Conditions:** The adoption of a variable returns to scale (VRS) model recognizes that students operate under different conditions and with different resources. This better reflects the reality of the educational context compared to constant returns to scale models.
- **Robustness of the Method:** DEA is less sensitive to outliers and non-normal distributions, making it suitable for educational data that often do not follow a normal distribution. This improves the robustness of our conclusions.
- **Identification of Areas for Improvement:** Efficiency scores below 1 indicate relative inefficiency, suggesting specific areas where students could improve. This is useful for designing targeted educational interventions that can help students optimize their use of soft skills to improve academic performance.
- **Significant Comparisons:** The results of Kruskal-Wallis analyses and pairwise comparisons showed significant differences in efficiency among different groups of students, highlighting the importance of considering demographic and behavioral variables in the design of educational programs.
- **Implications for Educational Policies:** Our analyses suggest that educational strategies should be adapted to meet the diverse needs of students. For example, younger students might benefit from greater support in soft skills, while older students might need different resources to make the most of their abilities.

In summary, DEA provided us with a detailed picture of student efficiency in the context of Serious Games, allowing us to identify areas of strength and improvement. The implications of these results are relevant for developing personalized educational interventions that can significantly improve overall academic performance (C. Zhou & Wang, 2009; H. Najadat et al., 2019).

5.6. Educational innovation and pedagogical contributions

This study introduces several groundbreaking innovations in educational assessment and academic guidance that significantly advance both educational technology and pedagogical practice. It presents the first comprehensive integration of serious games with machine learning algorithms specifically designed to predict academic pathways in higher education. This methodological innovation addresses three critical shortcomings of traditional assessment: it enables real-time behavioral analysis, provides transparent and interpretable predictive analytics, and supports holistic student profiling. Unlike conventional methods relying on static questionnaires or standardized tests, the serious game developed for this study captures authentic behavioral data during naturalistic decision-making scenarios. This allows educators to directly observe students' soft skills—such as time management, planning, and priority setting—in action, rather than depending on self-reported measures or artificial test environments.

The implementation of SHAP (SHapley Additive exPlanations) values represents another pedagogical advancement. By providing interpretable insights into the algorithm's reasoning, it allows educators to understand and communicate the rationale behind specific academic pathway recommendations. This level of transparency is essential in educational contexts where human oversight and student guidance remain crucial. Furthermore, this approach transcends traditional metrics of academic success by integrating soft skill performance with GPA, thereby generating comprehensive competency profiles that more accurately reflect students' real-world potential. The gamified environment also enables the collection of rich educational analytics, revealing learning behaviors that traditional assessments are unable to capture. These include micro-level patterns such as response times, strategic prioritization, and adaptive problem-solving—elements that reflect cognitive and behavioral traits relevant to academic and professional success.

In contrast to passive forms of assessment, the serious game maintains high engagement levels throughout the evaluation process. This results in more authentic performance data and reduced test anxiety. Moreover, the multi-dimensional structure of the assessment provides a broader understanding of each student's capabilities, encompassing far more than academic achievement alone. The implications for educational methodology and curriculum design are considerable. The predictive model supports the development of individualized learning pathways based on evidence-driven profiles rather than one-size-fits-all approaches. The identification of distinct soft skill patterns across different degree programs informs curriculum designers in aligning program requirements with students' actual competencies. Demographic analysis further reveals unique behavioral trends among students enrolled in various academic tracks, allowing for targeted interventions during critical stages of their academic journey. This research also proposes a new paradigm for academic advising and student support. Academic counselors can now supplement traditional metrics with behavioral evidence to offer more accurate and personalized guidance. Thanks to the automated nature of the game-based system, institutions can assess large student populations efficiently while preserving consistency and quality. The predictive capabilities of the model also support proactive guidance, helping students identify the most suitable academic paths before making potentially misaligned choices. Technologically, the study introduces several advances tailored to educational contexts. The serious game's cross-platform compatibility ensures accessibility for a diverse range of students. The use of automated scoring algorithms eliminates human bias and enhances reliability, while offering immediate, actionable feedback. Furthermore, the integration of explainable AI tools in this setting addresses a growing demand for transparency and accountability in algorithmic decision-making.

Overall, this research contributes to the evolution of educational assessment from static, standardized models to dynamic, interactive, and comprehensive systems. Its implications go beyond individual student evaluation, influencing institutional strategies, curricular frameworks, and policy development. By laying the groundwork for adaptive, personalized, and pedagogically sound technologies, the study supports a broader shift toward evidence-based practices that prepare students for increasingly complex educational and professional environments. In doing so, it establishes a robust framework for embedding behavioral analytics into decision-making processes within higher education, enabling a more nuanced understanding of student learning profiles and capabilities.

5.7. Strengths, limitations, and future research directions

One of the main strengths of the document is the comprehensive and detailed literature review covering the applications of serious games in various contexts, with an emphasis on educational benefits and integration with machine learning. This provides a solid theoretical framework that supports the relevance of the study. The research demonstrates significant contribution to educational theory and pedagogical practice through the integration of artificial intelligence in educational assessment, implementing multiple educational theories including constructivist learning principles through experiential game-based scenarios, Kolb's experiential learning cycle via realistic decision-making processes, and competency-based education frameworks through comprehensive soft skills evaluation. The methodology advances personalized learning pedagogy by creating individual competency profiles that extend beyond traditional academic metrics, supporting Gardner's Multiple Intelligences Theory through multi-dimensional capability assessment.

Additionally, the use of machine learning to evaluate student efficiency is methodologically rigorous and well-justified, demonstrating how these advanced techniques can be effectively applied in the educational context. This research establishes important theoretical foundations for AI-enhanced education by demonstrating how artificial intelligence can augment rather than replace human pedagogical judgment, with the integration of SHAP values creating a framework for transparent AI pedagogy that ensures algorithmic decisions remain interpretable and educationally meaningful. The detailed analysis of the predictive model's performance, using a range of metrics, further strengthens the validity of the results presented. The study contributes to emerging learning analytics theory by capturing authentic behavioral data during natural learning processes, providing insights into student capabilities that traditional assessment methods cannot access, supporting the development of predictive pedagogy approaches where AI enables proactive rather than reactive educational interventions based on evidence-based analysis of student competencies.

However, the study has some limitations that could affect the generalizability of the findings and their broader pedagogical applications. The sample size, consisting of 211 university students from a single university in Italy, is relatively limited for establishing universal pedagogical principles. Additionally, all participants were Computer Science degree students, which represents a significant limitation as this population may have specific characteristics regarding technology familiarity, problem-solving approaches, and soft skills patterns that may not be representative of students from other academic disciplines or educational contexts where AI applications might be implemented. Moreover, all participants had already completed the orientation phase and were enrolled in defined academic tracks, which limits the applicability of the findings to contexts where students are still in the process of choosing their educational pathway —a critical consideration for implementing AI-assisted academic guidance systems in real-world educational settings. Since admission to the programs in question is based on general merit criteria and does not involve any formal aptitude assessment, it cannot be assumed that students' academic choices are aligned with their actual transversal skill profiles. This reduces the potential of the model, in its current form, to serve as a predictive tool for academic guidance at earlier stages. This may not adequately represent the diversity of student populations in other regions or educational contexts, or academic fields. Furthermore, the study focuses on the use of serious games on smartphones and desktop PCs but does not consider other emerging educational technologies such as tablets, VR headsets, or augmented reality platforms, which could provide different user experiences and pedagogical outcomes. This limits the understanding of the effectiveness of AI-enhanced serious games across a broader range of educational technology ecosystems that are increasingly important in modern pedagogical environments. The inverse relationship between GPA and EBA classification observed in this study may be specific to the Italian higher education context and the particular characteristics of the sampled EBA program. Cross-cultural validation would be necessary to confirm the generalizability of this finding across different educational systems and pedagogical cultures, which is crucial for developing universally applicable AI in education frameworks.

To address these limitations and further deepen the knowledge in the field, several future research directions are necessary that will contribute to both technical development and pedagogical theory. First, it would be beneficial to conduct longitudinal studies to assess the long-term impact of AI-enhanced serious games on student learning outcomes, competency development, and academic success. This would provide crucial evidence for understanding how these AI-powered pedagogical interventions influence skill development over time and their effectiveness in supporting sustained learning and academic achievement. Such studies would contribute to theories of AI-mediated learning progression and help establish evidence-based guidelines for implementing AI systems in educational curricula. Second, exploring the application of serious games and machine learning techniques in different disciplines could provide further insights into their effectiveness in various educational contexts and pedagogical domains. This research would contribute to understanding how AI-enhanced assessment methods can be adapted to different learning objectives, disciplinary requirements, and pedagogical approaches, supporting the development of discipline-specific AI in education frameworks. Finally, investigating the use of advanced technological integrations, such as augmented reality (AR) and virtual reality (VR), and adaptive learning systems, could enhance the understanding of how different platforms influence learning outcomes and pedagogical effectiveness. This research direction would contribute to emerging theories of immersive learning environments and their integration with AI analytics for educational assessment and guidance. Future research should also focus on developing comprehensive frameworks for ethical AI implementation in educational settings, addressing issues of algorithmic bias, student privacy, data governance, and the preservation of human-centered pedagogical values. Research into effective methods for preparing educators to work with AI-enhanced educational tools represents a critical area for future investigation, including understanding how teachers can maintain their pedagogical expertise while effectively integrating AI systems into their practice, contributing to theories of human-AI collaboration in educational contexts.

Future research should aim to expand on these findings by exploring more diverse samples, integrating advanced technologies, and conducting longitudinal studies and addressing the ethical and pedagogical challenges associated with AI implementation in educational contexts. This comprehensive research agenda will further validate and enhance the application of serious games and AI analytics in educational settings while contributing to the theoretical foundations of AI-enhanced pedagogy and intelligent educational systems. This research establishes important precedents for how AI systems can be designed and implemented in educational contexts while preserving pedagogical integrity and supporting established learning theories, providing a foundation for future research in educational artificial intelligence and contributing to the growing body of knowledge about human-AI collaboration in learning environments.

6. Conclusions

In conclusion, this study highlights the significant potential of serious games combined with machine learning algorithms to evaluate and enhance student efficiency. The findings demonstrate that this game-based approach can effectively assess the relative efficiency of students by considering multiple inputs (soft skills) and outputs (GPA), providing a comprehensive understanding of their performance. The integration of machine learning techniques allows for the development of accurate predictive models that can inform personalized educational interventions.

Additionally, the use of Data Envelopment Analysis (DEA) as a complementary experiment provided further insights into student efficiency, identifying specific areas for improvement. DEA allowed for the comparison of the relative efficiency of students based on various inputs and outputs, offering a multidimensional view of their performance.

The results emphasize the importance of considering demographic and behavioral variables in educational strategies, suggesting that tailored support can significantly improve student outcomes. Beyond educational assessment, this research offers significant implications for career guidance as a professional activity. The serious games methodology demonstrates potential for transforming career counseling by providing objective, behavioral-based evidence of individual competencies that complement traditional career assessment tools. Career guidance counselors could leverage this approach to offer more personalized and evidence-informed recommendations, moving beyond subjective self-assessments or standardized aptitude tests. The machine learning models developed in this study could serve as decision-support tools for career professionals, helping them identify patterns between soft skills profiles and successful academic-career pathways. This data-driven approach to career guidance acknowledges that effective career selection inherently leads to more meaningful and successful educational experiences, as individuals are better matched to environments that align with their authentic competencies and behavioral strengths. The integration of serious games in career guidance practices could particularly benefit individuals who struggle with traditional assessment methods or have difficulty articulating their own strengths and preferences through conventional career exploration activities. Notably, the inverse relationship between GPA and EBA classification suggests that traditional academic metrics may not fully capture the competencies valued in business administration programs, highlighting the importance of holistic assessment approaches that include soft skills evaluation. Despite some limitations, such as the sample size and limited analysis of specific devices, the study provides valuable insights into the benefits of serious games and advanced analytics in education. Future research should expand on these findings by exploring more diverse samples, integrating advanced technologies, and conducting longitudinal studies to further validate and enhance the application of serious games in educational contexts.

CRediT authorship contribution statement

Agostino Marengo: Supervision, Methodology, Investigation, Conceptualization. **Alessandro Pagano:** Investigation, Formal analysis. **Vito Santamato:** Writing – review & editing, Writing – original draft, Investigation, Data curation.

Statements on open data and ethics

The study was submitted to the Ethics Committee of the University of Foggia for approval, and a decision is currently pending. All participants provided informed consent prior to their involvement, and their privacy rights were fully respected throughout the research process. The dataset generated and analyzed during the current study is not publicly available due to the ongoing ethical review; however, it may be obtained from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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